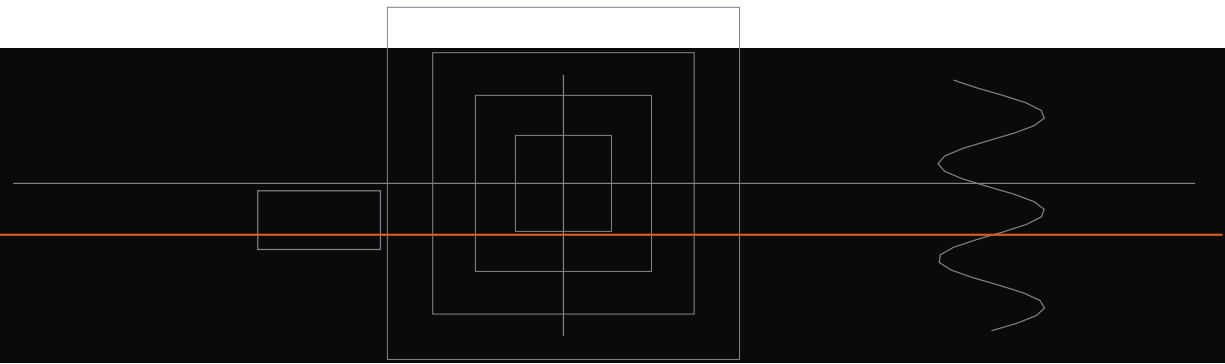


BILL OF MATERIALS

Quadruped II

Rev. 2026-04-30 96 items



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Product/Project: Quadruped II
Revision date: 2026-04-30

<https://www.printables.com/model/1451339-quadruped-ii>

ELECTRICAL COMPONENTS

22 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	0001-1202 Power_Dist-PCB	1	Order PCB HERE		Power Distrivbution PCB - PCB Files available for Clubmembers
>>	Servo Motors	x12	AliExpress Link	Amazon Link	180 degrees, 35kg (Approx 49.5x20,2x40mm)
->	ALT Servo Motors	x12	AliExpress Link	Amazon Link	Budget friendly alternative.
->	ALT Servo Motors	x12	AliExpress Link	Amazon Link	Budget friendly alternative.
>>	Raspberry Pi 4 Model B	1	AliExpress Link	Amazon Link	8gb Ram
>>	LiPo Battery	1	AliExpress Link	Amazon Link	4000mAh 4S 45~90C Lipo Pack w/XT-90
>>	IMU	1	AliExpress Link	Amazon Link	Integrated Measuring Unit
>>	Raspberry Pi Breakout Board	1	AliExpress Link	Amazon Link	Made by Dfrobot
>>	Fan	x2	AliExpress Link	Amazon Link	60x60 12v
>>	Servo Driver PCA9685	1	AliExpress Link	Amazon Link	Communication via I2C interface
>>	Arduino Nano	1	AliExpress Link	Amazon Link	For IMU, Battery sensing, LCD screen
>>	DC to DC (Buck) converter 20A	1	AliExpress Link	Amazon Link	Adjustable, set to ~7V
>>	DC to DC (buck) converter 3A	1	AliExpress Link	Amazon Link	For fans
>>	32GB MicroSD card	1	AliExpress Link	Amazon Link	32GB is maximum Pi supports
>>	RasbPi Cooling System	1	AliExpress Link	Amazon Link	-
>>	Arduino Nano Breakout Board	1	AliExpress Link	Amazon Link	-

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Product/Project: Quadrupe II
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<https://www.printables.com/model/1451339-quadrupe-ii>

ELECTRICAL COMPONENTS continued

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	GPIO Cable	1	AliExpress Link	Amazon Link	To connect Pi to its breakout board
>>	Logic Level Converter	1	AliExpress Link	Amazon Link	For adjusting voltage between the pi and the nano
>>	XT60H Connector	1	AliExpress Link	Amazon Link	To connect LiPo to the robot
>>	LCD Screen	1	AliExpress Link	Amazon Link	1.47 inch
>>	PS4 Controller	1	AliExpress Link	Amazon Link	-
>>	Digital Voltmeter	1	AliExpress Link	Amazon Link	DC 2.5-30V, used to measure battery voltage

BOM

Product/Project: Quadruped II
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<https://www.printables.com/model/1451339-quadruped-ii>

MECHANICAL COMPONENTS

28 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Bearing	x4	AliExpress Link	Amazon Link	6705-2RS 25X32X4
>>	Thrust Bearings	x4	AliExpress Link	Amazon Link	F4-10M 4x10x4mm
>>	Bearing	x20	AliExpress Link	Amazon Link	684ZZ Mini Deep Groove Ball Bearing 4x9x4mm
>>	Servo horns	x4	AliExpress Link	Amazon Link	RC Car Steering Servo Horn Arm 25T Teeth Long
>>	Heatsinks	x12	AliExpress Link	Amazon Link	6mm, black. Cut in half for each servo
>>	Servo Discs	x10	AliExpress Link	Amazon Link	with screws.
>>	Threaded rods - Long	x4	AliExpress Link	Amazon Link	M3 x 110mm (cut to length if needed)
>>	Threaded rods - Short	x4	AliExpress Link	Amazon Link	M3 x 25mm (cut to length if needed)
>>	Magnets	x8	AliExpress Link	Amazon Link	Used to hold fans in place
>>	Ballt Joint (Long joint)	x8	AliExpress Link	Amazon Link	S - 3mm L30mm M3
>>	Ball Join (Short Joint)	x8	AliExpress Link	Amazon Link	N - 3mm L20mm M3
>>	On/off Switch	1	AliExpress Link	Amazon Link	20A at 14VDC rated
>>	Spring	1	AliExpress Link	Amazon Link	For battery latch mechanism. 35mm long x 7mm OD
>>	Smooth Rod 3x80	1	AliExpress Link	Amazon Link	For battery hinge mechanism 3x75-80mm (cut to length if needed)
>>	Misc	1	AliExpress Link	Amazon Link	-

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MECHANICAL COMPONENTS continued

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Male Connector (for peripherals)	x12	AliExpress Link	Amazon Link	B2B-PH-K-S(LF)(SN)
>>	Socket for Connectors	x25	AliExpress Link	Amazon Link	SPH-002T-P0.5S
>>	Female Connector (for peripherals)	x12	AliExpress Link	Amazon Link	PHR-2
>>	Capacitor	1	AliExpress Link	Amazon Link	2200uF 16VDC Electrolytic RB Capacitor ,For servo board
>>	Reverse Polarity Protection Diode	1	AliExpress Link	Amazon Link	B160-13-F
>>	Ferrules	1	AliExpress Link	Amazon Link	Optional, if used need approx 20 across robot
>>	Terminal Blocks	x8	AliExpress Link	Amazon Link	1935776
>>	100k resistors	1	AliExpress Link	Amazon Link	For dividing the voltage at the buck converter so that it is readable by the nano
>>	47k Resistor	1	AliExpress Link	Amazon Link	CRGCQ1206F47K
>>	10k resistor	1	AliExpress Link	Amazon Link	CRGCQ0805F10K
>>	Heat Shrink	1	AliExpress Link	Amazon Link	Only small amount required, for covering exposed solid-core wires
>>	Power Delivery Wire	1	AliExpress Link	Amazon Link	Minimum 12AWG, for wires expected to carry current above 2A and/or voltage above 5V
>>	Connection Wire	1	AliExpress Link	Amazon Link	19-30AWG, recommend solid core, used for all other wires

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FASTENERS

22 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	MC6S-M2x8	x8	AliExpress Link	Amazon Link	M2 - Socket Head Cap Screw
>>	M6CS-M2x4	x14	AliExpress Link	Amazon Link	M2 - Socket Head Cap Screw
>>	MC6S-M2.5x6	x8	AliExpress Link	Amazon Link	M2.5 - Socket Head Cap Screw
>>	MC6S-M3x8H	x38	AliExpress Link	Amazon Link	M3 - Socket Head Cap Screw
>>	MC6S-M3x10H	x4	AliExpress Link	Amazon Link	M3 - Socket Head Cap Screw
>>	MC6S-M3x14H	x6	AliExpress Link	Amazon Link	M3 - Socket Head Cap Screw
>>	MC6S-M3x20H	x16	AliExpress Link	Amazon Link	M3 - Socket Head Cap Screw
>>	MC6S-M3x25	x4	AliExpress Link	Amazon Link	M3 - Socket Head Cap Screw
>>	MC6S-M4x25H	x4	AliExpress Link	Amazon Link	M4 - Socket Head Cap Screw
>>	MC6S-M4x16H	x4	AliExpress Link	Amazon Link	M4 - Socket Head Cap Screw
>>	MF6S-M3x8	x52	AliExpress Link	Amazon Link	M3 - Countersunk Socket Head Cap Screw
>>	K6S - M3x8	x4	AliExpress Link	Amazon Link	M3 - Socket Button Head Screw
>>	K6S - M3x10	x10	AliExpress Link	Amazon Link	M3 - Socket Button Head Screw
>>	K6S - M3x16	x36	AliExpress Link	Amazon Link	M3 - Socket Button Head Screw
>>	K6S - M3x20	x20	AliExpress Link	Amazon Link	M3 - Socket Button Head Screw
>>	RBS M3 (3,2x9x0,8))	x4	AliExpress Link	Amazon Link	M3 Washer with Large O.D
>>	LM6M_A4-70(M3)	x24	AliExpress Link	Amazon Link	M3 Lock Nut
>>	ML6M M3 Nut	x8	AliExpress Link	Amazon Link	M3 Nut - Low profile

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FASTENERS continued					
	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Heat insert M3 3x5	x96	AliExpress Link	Amazon Link	<i>L=3 O.D=5</i>
>>	Heat Insert M2 3x3	x18	AliExpress Link	Amazon Link	<i>L=3 O.D=3</i>
>>	Heat Insert M2.5 3x3.5	x8	AliExpress Link	Amazon Link	<i>L=3 O.D=3</i>
>>	Heat Insert M4 4x6	x8	AliExpress Link	Amazon Link	<i>L=4 O.D=6</i>

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ASSORTMENT BOXES 7 items					
	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Socket Button Head Screw	1	AliExpress Link	Amazon Link	M2-M3 x 2510pcs
>>	Socket head cap Screw	1	AliExpress Link	Amazon Link	2500pcs M2M2.5M3
>>	Socket head cap Screw	1	AliExpress Link	Amazon Link	1410pcs M3M4M5
>>	Countersunk Socket Head Cap Screw	1	AliExpress Link	Amazon Link	3045pcs - M2, M3, M4, M5, M6, M8
>>	Heat Insert (with press)	1	AliExpress Link	Amazon Link	1040pcs - M2-M4
>>	Heat Inserts	1	AliExpress Link	Amazon Link	1400pcs - M2-M4
>>	Lock Nut	1	AliExpress Link	Amazon Link	230pcs M3,M4,M5,M6,M8

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FILAMENT 4 items					
	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	PC / ASA / ABS / PETG Filament	1	AliExpress Link	Amazon Link	-
>>	PC-CF / PETG-CF / PA-CF / ABS+ (Engineering Grade Filaments)	1	AliExpress Link	Amazon Link	-
>>	PLA	1	AliExpress Link	Amazon Link	-
>>	TPU	1	AliExpress Link	Amazon Link	-

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TOOLS

13 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Soldering Iron TS101	1	AliExpress Link	Amazon Link	(Recommended)
>>	Heat Inset Tip Kit - For TS101	1	AliExpress Link	Amazon Link	(Recommended)
>>	Electric Precision Screwdriver	1	AliExpress Link	Amazon Link	(Recommended)
>>	Allen Key Set	1	AliExpress Link	Amazon Link	(Recommended)
>>	Torx Key Set	1	AliExpress Link	Amazon Link	(Recommended)
>>	Torque Screwdriver	1	AliExpress Link	Amazon Link	(Recommended)
>>	Wire stripper	1	AliExpress Link	Amazon Link	(Recommended)
>>	Wire cutter	1	AliExpress Link	Amazon Link	(Recommended)
>>	Heat Insert Press Tool	1	AliExpress Link	Amazon Link	(Recommended)
>>	Crimp Tool	1	AliExpress Link	Amazon Link	(Recommended)
->	ALT Soldering Iron - Generic	1	AliExpress Link	Amazon Link	Budget friendly alternative
->	ALT Heat insert tip - set	1	AliExpress Link	Amazon Link	Budget friendly alternative
>>	Precision Screwdriver Set	1	AliExpress Link	Amazon Link	Budget friendly alternative